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Wednesday 7:34 PM

Which of the following is most likely to increase the price of a forward contract on an asset? Higher dividends from a stock. ® ® ® ® B) C) Question #101 of 120 Question ID: 1207723 A) B) C) Question #102 of 120 Question ID: 1207727 A) B) C) Lower storage costs for a commodity. Lower convenience yield for a commodity

Yes — this is a very good CFA question because it tests the **cost-of-carry model** and especially the effect of **convenience yield**.

✅ Correct Answer: C. Lower convenience yield for a commodity

Let's examine each choice carefully.

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1. The basic forward pricing formula

For a commodity:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

where:

- F_0 = forward price
- S_0 = spot price
- r = financing/risk-free rate
- u = storage costs
- y = convenience yield
- T = time

The important part is:

$$\boxed{r + u - y}$$

Notice:

Storage cost is added.

Convenience yield is subtracted.

A. Higher dividends from a stock ❌

For a stock paying dividends:

$$\boxed{F_0 = (S_0 - PV(\text{dividends}))(1+r)^T}$$

Dividends are a **benefit to owning the stock**, so they reduce the forward price.

Example

Stock:

$$S_0 = \$100$$

Suppose the stock pays a dividend with PV of:

$$\$5$$

Then:

$$F_0 = (100 - 5)(1+r)^T$$

So:

$$F_0 = 95(1+r)^T$$

If dividends increase, the amount subtracted becomes larger.

Therefore:

$$\boxed{\text{Dividends} \uparrow \rightarrow \text{Forward price} \downarrow}$$

So A is ❌.

B. Lower storage costs for a commodity ❌

For commodities:

$$F_0 = S_0 e^{(r+u-y)T}$$

Storage costs u are part of the **cost of carrying the commodity**.

If storage costs decrease:

$$u \downarrow$$

then:

$$r + u - y \downarrow$$

Therefore:

$$\boxed{F_0 \downarrow}$$

Example

Suppose:

- Interest cost = 5%
- Storage cost = 3%
- Convenience yield = 1%

Net carry:

$$5\% + 3\% - 1\% = 7\%$$

Now storage costs fall to 1%:

$$5\% + 1\% - 1\% = 5\%$$

The forward price falls.

So B is ❌.

C. Lower convenience yield for a commodity ✅

This is the interesting one.

Remember:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

Convenience yield y is **subtracted**.

Therefore, if convenience yield falls:

$$y \downarrow$$

then:

$$r + u - y \uparrow$$

And therefore:

$$\boxed{F_0 \uparrow}$$

So:

✅ C is correct.

But WHY does lower convenience yield increase the forward price?

Think about what convenience yield means:

> **The benefit of physically owning the commodity today.**

Suppose you own physical crude oil.

If crude oil is extremely scarce, having it physically available is very valuable.

You can immediately:

- use it,
- process it,
- sell it,
- avoid production interruptions.

Therefore:

$$\boxed{\text{Scarcity} \uparrow \rightarrow \text{Convenience yield} \uparrow}$$

Now imagine the commodity is abundant.

You don't get much special benefit from physically holding it.

Therefore:

$$\boxed{\text{Convenience yield} \downarrow}$$

Since convenience yield is a benefit that offsets carrying costs, a lower convenience yield means:

$$\boxed{\text{Net cost of carry} \uparrow}$$

And therefore:

$$\boxed{\text{Forward price} \uparrow}$$

Let's use numbers

Suppose:

$$S_0 = \$100$$

$$r = 5\%$$

$$u = 3\%$$

Case 1: Convenience yield = 4%

$$r + u - y$$

$$= 5\% + 3\% - 4\%$$

$$= 4\%$$

So:

$$F_0 = 100e^{0.04T}$$

Case 2: Convenience yield falls to 1%

Now:

$$5\% + 3\% - 1\% = 7\%$$

So:

$$F_0 = 100e^{0.07T}$$

The forward price is now **higher**.

Why?

Because you've lost some of the benefit of physically holding the commodity.

🔥 The complete CFA relationship

Think of the commodity's forward price as:

$$\boxed{\text{Forward Price} = \text{Spot} + \text{Carrying Costs} - \text{Benefits}}$$

Specifically:

Costs:

$$\boxed{r + \text{storage costs}}$$

Benefit:

$$\boxed{\text{convenience yield}}$$

Therefore:

$$\boxed{F = Se^{(r+u-y)T}}$$

🧠 Memorize this table

Factor	Effect on Forward Price
Interest rate ↑	**Forward ↑**
Storage cost ↑	**Forward ↑**
Convenience yield ↑	**Forward ↓**
Dividends ↑	**Forward ↓**

So for your question:

A. Higher dividends ❌

Dividends are a benefit → forward price falls.

B. Lower storage costs ❌

Lower carrying cost → forward price falls.

C. Lower convenience yield ✅

Lower benefit from physical ownership → forward price rises.

$$\boxed{\textbf{C. Lower convenience yield}}$$

⭐ Super-short memory trick:

> **Costs ↑ → Forward ↑**
> **Benefits ↑ → Forward ↓**

Storage = **cost** → add it.

Convenience yield = **benefit** → subtract it.

Dividends = **benefit** → subtract them.

A bank borrows for 360 days and simultaneously lends the proceeds for 90 days. This transaction creates a synthetic forward rate agreement (FRA) closest to: a long position in a 90-day FRA on 270-day LIBOR. a long position in a 90-day FRA on 360-day LIBOR. a short position in a 360-day FRA on 90-day LIBOR.

Yes. This is a **synthetic FRA** question, and the wording is deliberately tricky. The key is to identify **what period the bank is effectively locking in**.

✅ Answer: A. Long position in a 90-day FRA on 270-day LIBOR

Let's break it down carefully.

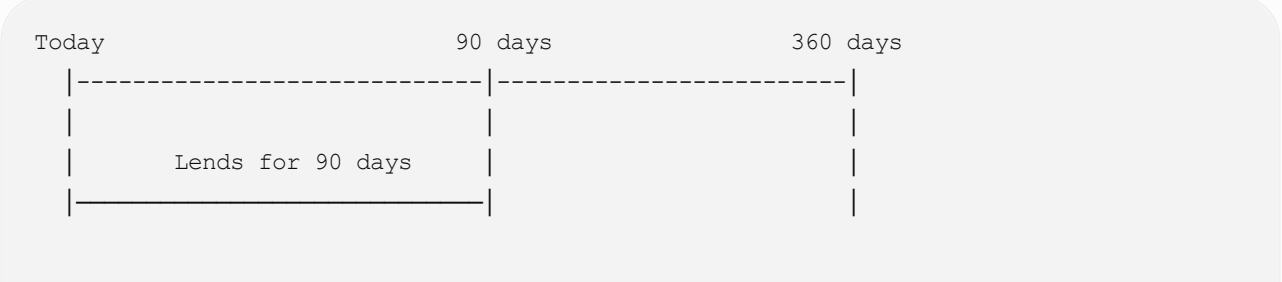
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1. What is the bank doing?

The bank:

> **Borrows for 360 days**
> and simultaneously
> **lends the proceeds for 90 days.**

Think of the timeline:



| | |

So after **90 days**, the bank's loan has matured, but its borrowing continues for another:

$$360 - 90 = \boxed{270 \text{ days}}$$

That is the key.

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2. What happens after 90 days?

Suppose the bank lends the money for 90 days.

After 90 days:

- The **loan to the borrower ends**.
- The bank gets its principal back.
- But the bank still owes money on its **360-day borrowing**.

So the bank will effectively need to **refinance/lend?** Let's focus on the interest-rate exposure.

The bank has locked itself into:

$$\boxed{90 \text{ days now} + \text{remaining } 270 \text{ days of borrowing}}$$

The future interest rate relevant to the remaining period is the rate applicable to the **270-day period beginning in 90 days**.

That's a **90-day forward-starting 270-day interest-rate exposure**.

---

3. What is an FRA?

An FRA is named based on:

$$\boxed{\text{length of the underlying interest period}}$$

and the time until that period begins.

Here:

- The future period begins in **90 days**
- The future borrowing period lasts **270 days**

Therefore:

$$\boxed{\text{90-day FRA on 270-day LIBOR}}$$

---

4. Is the bank LONG or SHORT?

This is the second part that students often get wrong.

The bank will need to **borrow/refinance money after 90 days**.

Therefore, the bank is worried about:

$$\boxed{\text{Interest rates rising}}$$

If rates rise, the bank's future borrowing becomes more expensive.

So the bank wants protection against **rising interest rates**.

Remember our FRA rule:

$$\boxed{\text{Long FRA gains when reference rate rises}}$$

Therefore the bank wants:

$$\boxed{\text{LONG FRA}}$$

So:

✅ A. Long position in a 90-day FRA on 270-day LIBOR

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5. Let's use a numerical example

Suppose the bank does this:

Today

Bank:

- Borrows $10 million for **360 days**
- Lends $10 million for **90 days**

After 90 days, the bank receives its $10 million back from the borrower.

But the bank's borrowing continues for another:

$$360 - 90 = 270 \text{ days}$$

Suppose the interest rate after 90 days suddenly rises.

Today:

Expected 270-day rate = **5%**

After 90 days:

Actual 270-day LIBOR = **8%**

The bank now faces a much higher cost of borrowing for those remaining 270 days.

That's exactly the risk the **long FRA** protects against.

The FRA effectively gives the bank compensation when:

$$\text{270-day LIBOR} > \text{FRA fixed rate}$$

---

6. Why isn't B correct?

B. Long position in a 90-day FRA on 360-day LIBOR ❌

The problem is the **360-day LIBOR**.

The bank's original borrowing lasts 360 days, but after the first 90 days, only:

$$270 \text{ days}$$

remain.

The relevant future interest period is therefore **270 days**, not 360 days.

So:

$$\boxed{\text{90-day FRA on 270-day LIBOR}}$$

not:

$$\text{90-day FRA on 360-day LIBOR}$$

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7. Why isn't C correct?

C. Short position in a 360-day FRA on 90-day LIBOR ❌

This reverses both the timing and the rate period.

A short FRA benefits when interest rates **fall**.

But the bank is exposed to:

$$\boxed{\text{rates rising}}$$

because it will need to continue borrowing.

Therefore, it needs a:

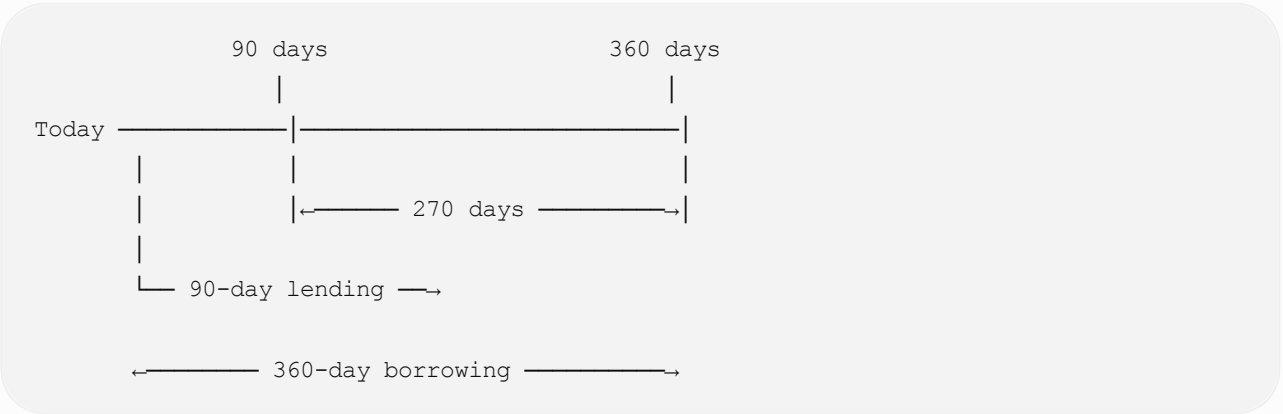
$$\boxed{\text{LONG FRA}}$$

not a short FRA.

🧠 The most important CFA trick

When you see a question like this, **draw the timeline**.

Here:



Then ask:

Step 1: When does the future exposure begin?

$$\boxed{90 \text{ days}}$$

Step 2: How long does the future exposure last?

$$360 - 90 = \boxed{270 \text{ days}}$$

Therefore:

$$\boxed{\text{90-day FRA on 270-day LIBOR}}$$

Step 3: Is the bank afraid of rates rising or falling?

It's a future **borrower** → afraid of:

$$\boxed{\text{Rates} \uparrow}$$

Step 4: Who gains when rates rise?

$$\boxed{\text{Long FRA}}$$

Therefore:

🎯 A. Long position in a 90-day FRA on 270-day LIBOR

Memorize this pattern:

> **Future borrowing → Long FRA → protects against rising rates.**

And:

> **Future lending → Short FRA → protects against falling rates.**

A characteristic that distinguishes derivatives from ETFs is that derivatives: transform the performance of their underlying assets. are contingent claims on the performance of their underlying assets. derive their performance from the performance of their underlying assets.

The correct answer is:

> **Derivatives are contingent claims on the performance of their underlying assets.** ✅

Let's understand the difference

The key distinction is **how the investment's value is linked to the underlying asset**.

	Derivatives	**ETFs**
What are they?	Contracts	Investment funds
Value depends on underlying?	✅ Yes	✅ Usually
Contingent claim?	✅ Yes	❌ No
Investor owns a claim/contract whose payoff depends on an	✅	❌

	Derivatives	ETFs
underlying event/value?		
Example	Call option on Apple	ETF holding Apple shares

Why "contingent claim"?

A **derivative's payoff is contingent on what happens to the underlying asset**.

For example, suppose you buy a call option:

- Apple stock = $100
- Call strike = $105
- At expiration Apple = $120

Your call payoff:

$$\max(120 - 105, 0) = \$15$$

If Apple finishes at $90:

$$\max(90 - 105, 0) = 0$$

So the derivative's payoff is **contingent upon the performance of Apple stock**.

Why the other choices are wrong

A. "Derivatives transform the performance of their underlying assets." ❌

This is not the defining characteristic.

A derivative can **alter or transform exposure**, but that's not what distinguishes it from an ETF.

For example, a put option can provide downside protection, but "transform" isn't the key CFA definition.

C. "Derivatives derive their performance from the performance of their underlying assets." ❌

This sounds very tempting!

In fact, **ETFs can also derive their performance from underlying assets**.

For example, an S&P 500 ETF attempts to provide returns based on the performance of the S&P 500.

Therefore, this statement **doesn't distinguish derivatives from ETFs**.

🎯 CFA Exam Shortcut

Remember:

> **Derivative = contingent claim**

A derivative's payoff is **conditional/contingent on the value or performance of an underlying asset, rate, index, etc.**

Examples:

- **Option** → payoff depends on underlying price
- **Futures** → payoff depends on underlying price
- **Forward** → payoff depends on underlying price
- **Swap** → payoff depends on underlying rate/price/index

One-line memory trick

ETF = owns/invests in assets
Derivative = contract whose payoff is contingent on an underlying

So the answer is **B — contingent claims on the performance of their underlying assets.**

> Derivatives typically transform the performance of their underlying assets, in contrast with passing the performance of the underlying assets through to the investor in the manner of mutual funds or ETFs. Mutual funds and ETFs, as well as derivatives, derive their performance from the performance of their underlying assets. Derivatives may be contingent claims or forward commitments.

Exactly — **the explanation you pasted is the correct CFA interpretation**, and it also clarifies why the earlier question can be confusing.

The key distinction

There are **two different ideas**:

1. **Derive performance from the underlying** → true for **both derivatives and ETFs**
2. **Transform the performance of the underlying** → characteristic of **derivatives**

So the question asks what **distinguishes derivatives from ETFs**.

Think of it this way

Suppose the underlying asset increases by **10%**.

ETF → passes the performance through
If you own an ETF tracking the asset:

> Underlying +10% → ETF approximately +10%

The ETF is essentially **passing through** the underlying performance.

Derivative → can transform the performance
Suppose you buy a call option.

Underlying:

> Stock goes from $100 → $110 = **+10%**

But your call might go from $5 → $8.

That's:

$$\frac{8-5}{5} = 60\%$$

So:

> Underlying +10% → Derivative +60%

The derivative has **transformed** the underlying's performance because of leverage, payoff structure, strike price, etc.

Now look at the three statements

Statement	Correct?	Why?
Derivatives transform the performance of underlying assets	✅	This distinguishes derivatives

Statement	Correct?	Why?
Derivatives derive performance from underlying assets	✅	But ETFs also do this
Derivatives are contingent claims	⚠️	Some derivatives are contingent claims, but some are **forward commitments**

This last point is important.

Derivatives have TWO broad forms

1. Contingent claim

Payoff depends on whether/how an event occurs.

Examples:

- Call option
- Put option

2. Forward commitment

Both parties are committed to perform in the future.

Examples:

- Forward
- Futures
- Swap

Therefore, saying **"derivatives are contingent claims"** is too narrow.

🎯 CFA exam rule

When you see:

> **Derivative vs. mutual fund/ETF**

Think:

ETF / Mutual Fund
→ **passes through** underlying performance

Derivative
→ **transforms** underlying performance

And remember:

> **All derivatives derive their performance from an underlying, but not all derivatives are contingent claims.**

That's why **"transform the performance"** is the best distinguishing characteristic.

can you help me with example ?

Absolutely. Let's use **one stock** and compare an **ETF vs. a derivative**. This makes the CFA wording "pass through" vs. "transform" very clear.

Example: Stock price rises from $100 → $110

Suppose the underlying stock increases by **10%**.

1. ETF — passes the performance through

Imagine an ETF that holds this stock.

You invest **$1,000**.

If the stock rises 10%, approximately:

$$\$1,000 \times 1.10 = \$1,100$$

So your return is:

$$\boxed{+10\%}$$

The ETF essentially **passes the underlying performance through to you**.

> Stock +10% → ETF approximately +10%

---

2. Derivative — transforms the performance

Now suppose instead you buy a **call option** on the same stock.

Stock:

- Current price = $100
- Call strike price = $100
- Option premium = $5

You pay only:

$$\$5$$

for the option.

Suppose the stock rises to **$110**.

The call's intrinsic value becomes:

$$\$110 - \$100 = \$10$$

You paid $5 and now have an option worth $10.

Your return is:

$$\frac{10 - 5}{5} = 100\%$$

So:

> Stock **+10%** → Call option **+100%**

That's what CFA means by **"transform the performance."**

The derivative did **not simply give you the same 10% return**. Its payoff structure transformed the underlying's movement into a different return.

Why does this happen?

Because a derivative has a **contractual payoff formula**.

For a call:

$$\text{Payoff} = \max(S_T - X, 0)$$

So if:

Stock ends at $90

$$\max(90 - 100, 0) = 0$$

Call payoff = $0.

Stock ends at $110

$$\max(110 - 100, 0) = 10$$

Call payoff = $10.

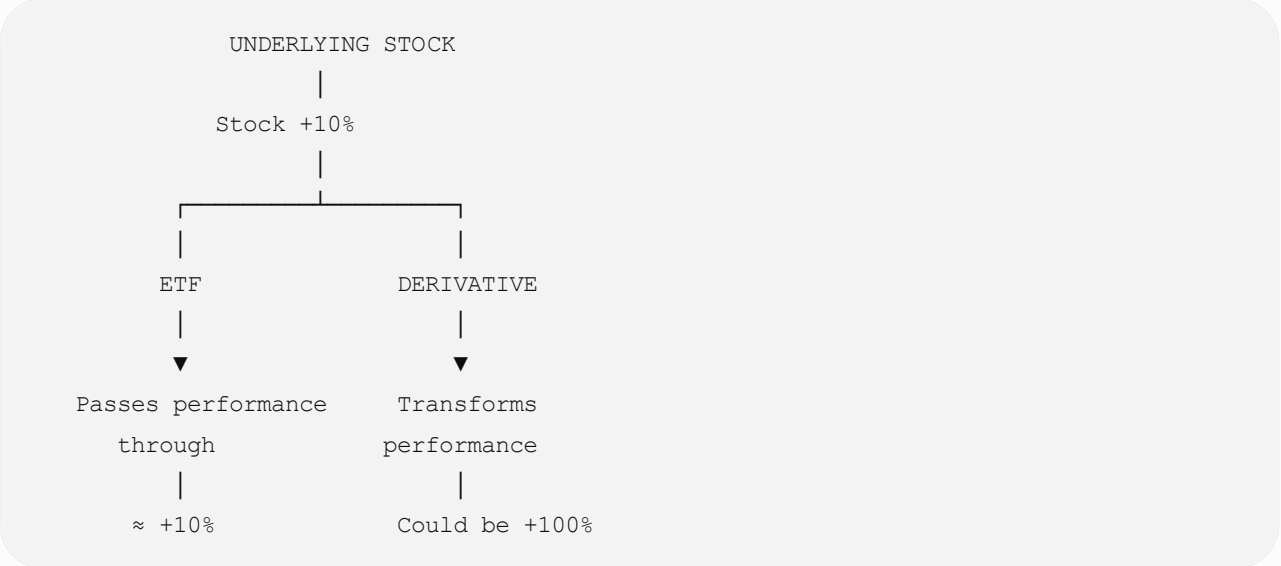
Stock ends at $130

$$\max(130 - 100, 0) = 30$$

Call payoff = $30.

The option **changes the way you experience the underlying's performance**.

The CFA concept in one picture



Another very simple example

Imagine the stock goes:

$$\$100 \rightarrow \$120$$

Stock return:

$$\frac{120 - 100}{100} = 20\%$$

ETF: approximately **+20%**

Call option: could go from $5 → $25, giving:

$$\frac{25 - 5}{5} = 400\%$$

So the underlying gained **20%**, but the derivative generated **400%**.

That's the **transformation**.

🎯 Remember this for the CFA exam

> **ETF / Mutual Fund → "Pass through"**
>
> **Derivative → "Transform"**

And don't confuse this with:

> **"Derive performance from the underlying."**

Both ETFs **and** derivatives derive their performance from an underlying.

The word that **distinguishes** derivatives is:

$$\boxed{\textbf{TRANSFORM}}$$

Also remember: **derivatives can be either contingent claims (options) or forward commitments (forwards, futures, swaps).**

A programming error resulted in the issuance of a forward contract that ignored the convenience yield investors realize from holding the underlying asset. To gain an arbitrage profit from this error, an arbitrageur's position would most likely include a long position in the: forward contract. underlying asset. risk-free asset.

Yes. This is a **classic cost-of-carry arbitrage** question. The key is understanding what happens when the forward contract **ignores convenience yield**.

Step 1: Start with the correct commodity forward formula

For a commodity:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

where:

- S_0 = spot price
- r = financing cost
- u = storage/carrying cost

- y = convenience yield

The important part:

$$\boxed{\text{Forward price} = \text{Spot} + \text{carrying costs} - \text{convenience yield}}$$

Step 2: What happens if the contract ignores convenience yield?

If the programmer **ignores** y, the system calculates:

$$F_{\text{wrong}} = S_0 e^{(r+u)T}$$

instead of:

$$F_{\text{correct}} = S_0 e^{(r+u-y)T}$$

Because y is subtracted, ignoring it makes the calculated forward price **too high**.

So:

$$\boxed{F_{\text{wrong}} > F_{\text{correct}}}$$

Step 3: What should the arbitrageur do?

If the forward is **overpriced**, you want to:

$$\boxed{\text{SHORT the forward}}$$

But the question asks which position would be included:

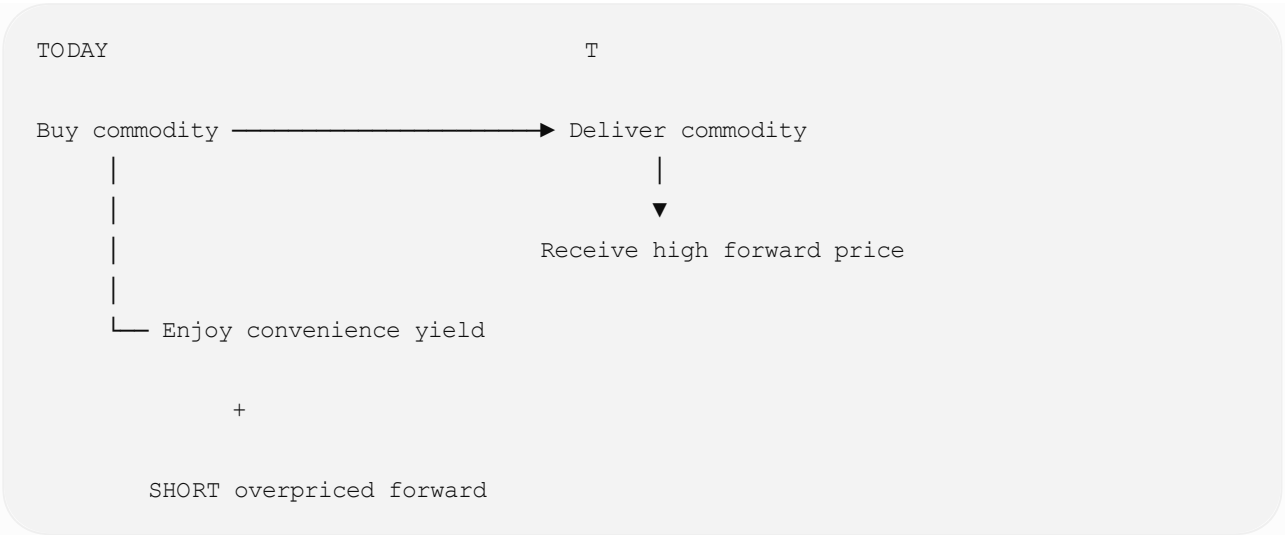
- Long forward ❌
- **Long underlying asset** ✅
- Long risk-free asset ❌

Why **long the underlying**?

Because the arbitrageur can:

1. **Buy the commodity today**
2. Hold it until the forward expiration
3. Receive the **convenience yield**
4. Deliver the commodity into the overpriced forward
5. Receive the inflated forward price

So the arbitrage strategy looks like:



🎯 Answer

$$\boxed{\textbf{B. Underlying asset}}$$

Why not the risk-free asset?

The risk-free asset is part of the **financing/carrying** side of the arbitrage, but the crucial mistake is that the forward **failed to account for the benefit of physically holding the commodity**.

You therefore need to **own the underlying** to capture that convenience yield.

CFA memory rule

Think:

> **Convenience yield = benefit from owning the physical commodity.**

Therefore:

Forward ignores convenience yield → forward price too high → short forward + long commodity.

And the broader formula to memorize:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

Costs go UP → forward price ↑

Benefits from ownership (convenience yield) go UP → forward price ↓

can you help me with example

Absolutely. Let's use a **simple commodity example** so you can see exactly why the arbitrageur goes **long the underlying**.

Example: Gold with a convenience yield

Suppose today:

- Gold spot price = **$1,000**
- Risk-free rate = **5%**
- Storage cost = **2%**
- Convenience yield = **3%**
- Time = 1 year

For simplicity, we'll use simple compounding rather than continuous compounding.

1. Correct forward price

The basic relationship is:

$$F_0 = S_0[1 + r + u - y]$$

Substitute:

$$F_0 = 1000[1 + 0.05 + 0.02 - 0.03]$$

$$F_0 = 1000(1.04)$$

$$\boxed{F_0 = \$1,040}$$

So the **correct forward price is $1,040**.

Now the programming error

The computer **forgot the 3% convenience yield**.

So it calculates:

$$F_{\text{wrong}} = 1000[1 + 0.05 + 0.02]$$

$$= 1000(1.07)$$

$$\boxed{\$1,070}$$

The forward contract is therefore offering a price of **$1,070**, even though the fair price should be only **$1,040**.

So:

$$\boxed{\text{Forward is overpriced by } \$30}$$

What does the arbitrageur do?

The arbitrageur thinks:

> "This forward lets me sell gold for $1,070 when the fair forward price is only $1,040. Great! I'll lock in that high selling price."

Therefore:

Position 1: Long the underlying gold ✅

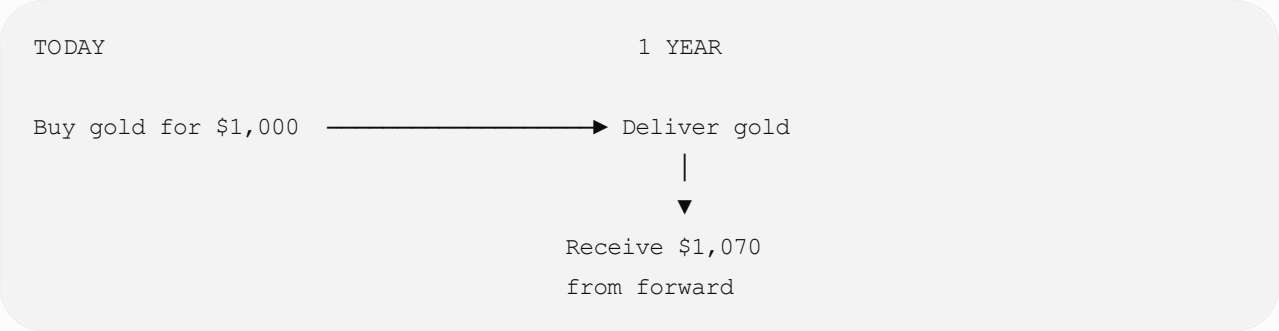
Buy gold today for $1,000.

Position 2: Short the forward ✅

Agree today to sell the gold in one year for **$1,070**.

Then hold the gold until expiration and deliver it into the forward.

Timeline



But there's something very important happening while you're holding the gold:

You receive the convenience yield.

The physical gold gives you a benefit that someone holding only a forward contract doesn't get.

That's why the **true cost of carrying the gold is lower** than the computer thinks.

Where is the arbitrage profit?

The market is effectively saying:

> "You can obtain gold today and carry it to the future for a cost consistent with $1,040."

But the erroneous forward lets you sell it for:

$$\$1,070$$

So there is an apparent locked-in difference of:

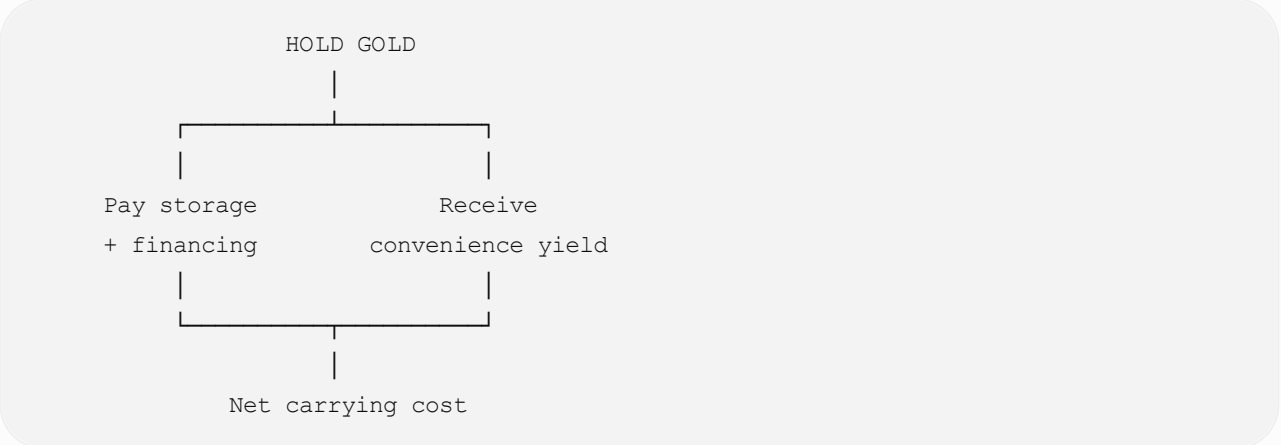
$$1070 - 1040 = \boxed{\$30}$$

The exact arbitrage profit calculation in a real problem would account for financing and storage cash flows precisely, but the important CFA concept is the direction.

Why LONG the underlying?

This is the part I want you to remember.

The **convenience yield only comes from physically owning the commodity**.



If you **don't own the gold**, you don't receive the convenience yield.

Therefore, to exploit a forward that **ignored convenience yield**, you need:

$$\boxed{\text{LONG underlying}}$$

and

$$\boxed{\text{SHORT forward}}$$

🎯 CFA exam shortcut

Whenever you see:

> **"Forward price is too high"**

Think:

$$\boxed{\text{Buy underlying} + \text{Short forward}}$$

Whenever you see:

> **"Forward price is too low"**

Think:

$$\boxed{\text{Short underlying} + \text{Long forward}}$$

And specifically for convenience yield:

$$\boxed{F = Se^{(r+u-y)T}}$$

Convenience yield ↑ → Forward price ↓

Convenience yield ignored → Calculated forward price too high → Short forward + Long underlying.

That is exactly why the answer to your question is **underlying asset**.

If the risk-free rate of interest is 3%, the spot price of an asset is 35, and its net cost of carry is –2, the no-arbitrage price of a one-year forward contract on units of the asset is closest to:

Yes. This is a **cost-of-carry** question. The important part is understanding what **net cost of carry = –2%** means.

Given

- Spot price: $S_0 = 35$
- Risk-free rate: $r = 3\%$
- Net cost of carry: $c = -2\%$
- Time: $T = 1$ year

The forward pricing formula is:

$$\boxed{F_0 = S_0(1 + r + c)^T}$$

Here, net cost of carry is **negative 2%**, so:

$$F_0 = 35(1 + 0.03 - 0.02)$$

$$F_0 = 35(1.01)$$

$$\boxed{F_0 = \$35.35}$$

🎯 **Answer: $35.35**

But why is the net cost of carry negative?

This is the key concept.

Remember:

$$\boxed{\text{Net cost of carry} = \text{Carrying costs} - \text{Benefits}}$$

For a commodity, for example:

$$\text{Net carry} = r + \text{storage cost} - \text{convenience yield}$$

So if net carry is **–2%**, it means the **benefits of holding the asset exceed the costs by 2%**.

That's why the forward price is only slightly above the spot price:

```
Spot price              $35.00
     |
     | + 3% financing
     | - 2% net benefit
     ▼
Forward price           $35.35
```

CFA shortcut

If you're given **net cost of carry directly**:

$$\boxed{F_0 = S_0(1 + r + \text{net cost of carry})^T}$$

So always **add the net cost of carry to the risk-free rate**.

Here:

$$3\% + (-2\%) = \boxed{1\%}$$

Therefore:

$$35 \times 1.01 = \boxed{35.35}$$

Answer = $35.35.

Why yuo have not add E to the power ?

Great catch. **You absolutely can see an $e^{(\cdot)}$ formula for forward pricing.** The difference is **how the interest/carry rates are quoted**.

Two versions of the formula

1. Continuous compounding

If the rates are continuously compounded:

$$\boxed{F_0 = S_0 e^{(r+c)T}}$$

For this question:

$$F_0 = 35e^{(0.03-0.02)(1)}$$

$$= 35e^{0.01}$$

$$\approx \boxed{35.35}$$

So **even using e, you get approximately $35.35**.

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2. Simple annual compounding

If the problem uses ordinary/simple annual compounding:

$$\boxed{F_0 = S_0(1 + r + c)^T}$$

Then:

$$F_0 = 35(1 + 0.03 - 0.02)$$

$$= 35(1.01) = \boxed{35.35}$$

Why do both give almost the same answer?

Because:

$$e^{0.01} = 1.01005$$

while:

$$1.01 = 1.01000$$

So:

$$35e^{0.01} = 35.3518$$

which rounds to:

$$\boxed{35.35}$$

🎯 CFA exam point

The **more general continuous-compounding cost-of-carry formula** is:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

where:

- r = financing cost
- u = storage/carrying cost
- y = convenience yield

If the question gives **net cost of carry** c:

$$\boxed{F_0 = S_0 e^{(r+c)T}}$$

So for your question, using e is perfectly valid:

$$\boxed{35e^{(0.03-0.02)(1)} \approx \$35.35}$$

Your instinct about the e was correct. I should have used the continuous-compounding version first, since that's the standard CFA cost-of-carry formulation.

ChatGPT can make mistakes. Check important info.